reality and equivalence in algebraic quantum theory

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overview

1. preliminaries
2. inequivalent representations
3. equivalent QFTs
4. same physics on all spacetimes

what does equivalence have to do with reality?

mantra: reality is invariant relative to equivalence of true descriptions of the world.

what sorts of things can be equivalent?

- sentences
- descriptions
- theories
  - sentences?
  - models?
  - functors?
- representations
- Lagrangians

inequivalent quantizations

OBJECTIVE: describe a quantum system corresponding to a classical system with configuration space X.

OBSTACLE: embarrassment of riches — if X is not simply connected, then there are inequivalent quantum theories corresponding to X.

from quantization to representation

- quantization: given configuration space X, find an corresponding quantum theory (hilbert space and operators satisfying certain algebraic relations).
- Landsman's maneuver: there is a unique C^*-algebra $A[X]$ such that Hilbert space representations of $A[X]$ are in one-to-one correspondence with quantizations of X.

algebraic imperialism (I.E. Segal)

"The physical content of the theory is completely contained in the abstract algebra."

Observables: self-adjoint operators in A.

States: positive, linear, normalized maps from observables to $\mathbb{R}$.

Dynamics: one-parameter group of automorphisms of A.

hilbert space conservatism

Representation: a module $\pi : A \times H \to H$

Fact: A representation $\pi : A \times H \to H$ induces a map $\pi^* : T_1(H) \to S(A)$, given by

$$\pi^*(d)(a) = \mathrm{Tr}(d\pi(a)).$$

Gelfand-Naimark-Segal Theorem: $\forall \omega \in S(A)$, there is a Hilbert space H_ω, an operator $d \in T_1(H_\omega)$, and a representation $\pi_\omega : A \times H_\omega \to H_\omega$ such that $\omega = \pi_\omega^*(d)$.

unitarily equivalent representations

A unitary equivalence $w : (H, \pi) \to (K, \phi)$ is a unitary map $w : H \to K$ that makes the following diagram commute:

$$\begin{array}{ccc} A \times H & \xrightarrow{\pi} & H \\ {\scriptstyle 1\times w}\downarrow & & \downarrow{\scriptstyle w} \\ A \times K & \xrightarrow[\phi]{} & K \end{array}$$

disregarding inequivalent representations

Fell's theorem: for any $d \in T_1(H)$, operators $a_1, \ldots, a_n \in A$, and $\epsilon > 0$, there is a $d' \in T_1(K)$ such that

$$\left|\mathrm{Tr}(d\pi(a_i)) - \mathrm{Tr}(d'\phi(a_i))\right| < \epsilon.$$

Claim (Segal): any two representations of $A[X]$ are physically equivalent.

regarding inequivalent representations

Claim (Ruetsche): physically equivalent representations are unitarily equivalent.

$\Rightarrow$ for non simply connected X, there are physically inequivalent representations of $A[X]$.

what's the issue?

1. hilbert space conservatism: there is one correct representation

   $\Rightarrow$ choosing a representation involves a further theoretical commitment

2. algebraic imperialism: all physical information is in the abstract algebra of observables

   $\Rightarrow$ choosing a representation entails no further theoretical commitment

problems for hilbert space conservatism

- underdetermination: what reason for choosing a representation?
- what is gained by choosing a representation?
- what is lost by choosing a representation?
- what kind of epistemic commitment is it to choose a representation?

Baker's conundrum

1. physical equivalence $\Leftrightarrow$ intertranslatability (Poincaré, Glymour, Quine, etc.)
2. fact: there are inequivalent representations that are intertranslatable.
3. conclusion: there are inequivalent representations that are physically equivalent.

Buridan's Ass meets QFT

1. The system can be represented by π or by ϕ, but not by both.
2. If the world is represented by π, it could equally well be represented by ϕ.

the dilemma restated

1. break the symmetry between the two representations
2. deny that there is any choice to be made

common theme: "freedom to choose a representation" is theoretically insignificant

equivalent nets

Let $\mathbb{K}$ be a family of "nice" subsets of Minkowski spacetime. A **net of observable algebras** is a map $A : \mathbb{K} \to \textbf{CStar}$ satisfying:

- isotony: if $O_1 \subseteq O_2$ then $A(O_1) \subseteq A(O_2)$
- microcausality: if $O_1 \perp O_2$ then $[A(O_1), A(O_2)] = \{0\}$

equivalent nets

Let $A, B : \mathbb{K} \to$ **CStar** be nets of observable algebras over Minkowski spacetime. A morphism $\alpha : A \to B$ is defined as a family $\alpha_O : A(O) \to B(O)$ of $*$-homomorphisms such that for any $O_1 \subseteq O_2$, the following diagram commutes:

$$\begin{array}{ccc} A(O_2) & \xrightarrow{\alpha_{O_2}} & B(O_2) \\ \uparrow & & \uparrow \\ A(O_1) & \xrightarrow[\alpha_{O_1}]{} & B(O_1) \end{array}$$

equivalent nets

question: how much information about **dynamics** is contained in the net?

example: let A_m be the net for the free Bose field of mass $m > 0$. Are A_m and $A_{m'}$ equivalent nets when $m \neq m'$?

(It is known that $A_m(O) \cong A_{m'}(O)$ for all O. "No" would indicate that the net contains information about the dynamics.)

locally covariant QFT

Let **Loc** be the category whose objects are globally hyperbolic manifolds, with Lorentzian metric, orientation, and time orientation; and whose arrows are causally convex embeddings.

A locally covariant QFT is a covariant functor A from **Loc** to **CStar**.

equivalent QFTs

definition: a morphism of locally covariant QFTs is a natural transformation $\alpha : A \to B$. This morphism is called an equivalence just in case each α_M is an isomorphism.

equivalent QFTs

question: suppose A, B are locally covariant QFTs such that $A(M) \cong B(M)$, for all spacetimes M. Does it follow that A and B are equivalent?

answer: no (Fewster)

same physics on all spacetimes

constructive QFT: same construction recipe, different background spacetimes

reverse engineering: given a locally covariant QFT A, when should we say that A describe the "same physics" on all spacetimes?

same physics on all spacetimes

weak necessary condition: for two QFTs A, B, if both describe the same physics on all spacetimes, and $A(M) \cong B(M)$ for some spacetime M, then $A \cong B$.